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Dark mode vs light mode in digital interfaces: Effects on subjective readability and cognitive load

A mixed-method experimental study of dark and light interface themes in content-heavy digital environments

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Abstract

In recent years, the use of dark mode and light mode color themes in digital interface designs has increased immensely, influencing usability, visual comfort and the user experience.

This study aims to investigate dark mode and light mode in digital interfaces and their effects on subjective readability and cognitive load. A mixed methods approach was used throughout this study, combining a within-subject experiment with a questionnaire consisting of Likert-scale questions together with open-ended questions. A total of 22 participants were involved and tested both prototypes followed by the questionnaire.

The results gathered through the experiments showed minor differences between the two prototypes in terms of subjective readability and cognitive load. Both the quantitative and qualitative results displayed similar ratings of subjective readability and cognitive load across each of the two interfaces. Overall the results indicate that both light mode and dark mode can be effective when the interface design is well developed, however users' individual preferences may influence the perceived experience.

The findings in this study provide information to future researchers and designers in the field of user interface design.

Keywords: *Dark Mode, Light Mode, Subjective readability, Cognitive load, Digital interface, User Experience*.

Table of Contents

Abstract **2**
1 Introduction 3
1.1 Problem statement 3
1.2 Purpose and research questions 4
1.3 Scope and delimitations 5
1.4 Disposition 6
2 Theoretical framework **7**
2.1 Dark Mode and Light Mode in Digital Interfaces 7
2.2 Readability in Digital Interfaces 7
2.3 Subjective readability 8
2.4 Cognitive Load Theory 9
2.5 User Experience and User Control 10
2.6 Summary of the theoretical framework 10
2.7 Conceptual definitions 11
Subjective readability 11
Cognitive load 11
Extraneous cognitive load 11
Visual comfort 11
User experience 12
Usability 12
3 Method and implementation **12**
3.1 Research design 12
3.2 Participants 13
3.3 Materials 13
3.4 Prototype design 14
3.5 Survey design 15
3.7 Task questions 17
3.8 Survey questions 17
3.9 Data collection 18
3.10 Data analysis 19
3.11 Validity and reliability 20
3.12 Considerations 21
4 Results **22**
4.1 Analysis of Likert-scale questions 23
4.2 Paired comparisons (t-test) 23
4.3 Correlation analysis 24
5 Discussion **31**
5.1 Research Question 1 31
5.2 Research Question 2 32
5.3 Method discussion 34

5.4 Methodological insights..... 34
5.5 Methodological limitations.....35
5.6 Reliability and validity reflection.....36
5.7 Methodological improvements.....37
6 Conclusions and further research.....38
6.1 Conclusions..... 38
6.2 Further research.....38
7 Reference.....40
8 Appendixes.....43
Appendix A - Dark mode prototype.....43
Appendix B - Light mode prototype.....45
Appendix C - Consent form.....47
Appendix D - Participants' Likert-scale responses for readability items in dark mode, including participant means, overall mean and standard deviation.....48
Appendix E - Participants' Likert-scale responses for cognitive load items in dark mode, including participant means, overall mean and standard deviation.....48
Appendix F - Participants' Likert-scale responses for readability items in light mode, including participant means, overall mean and standard deviation.....49
Appendix G - Participants' Likert-scale responses for cognitive load items in light mode, including participant means, overall mean and standard deviation.....49

List of figures

Figure 1: Primary color used in the prototypes..... 15
Figure 2: Mean and Standard deviation of subjective readability in light mode and dark mode..... 23
Figure 3: Mean and Standard deviation of cognitive load in light mode and dark mode.. 23
Figure 4: Results of t-tests.....24
Figure 5: Mean of subjective readability and cognitive load in dark mode..... 24
Figure 6: Scatter diagram with a fitted trendline for subjective readability and cognitive load in dark mode..... 25
Figure 7: Mean of subjective readability and cognitive load in light mode.....26
Figure 8: Scatter diagram with a fitted trendline for subjective readability and cognitive load in light mode.....26

1 Introduction

Dark mode and light mode are widely available design options in digital interfaces, including websites, mobile applications, and reading platforms. While both themes are extensively used, there is ongoing discussion regarding how they influence users' subjective readability (the perceived ease of comprehending text and content), cognitive load (mental effort), and overall user experience.

Previous research within human-computer interaction has examined how light mode (dark text on a light background) and dark mode (light text on a dark background) affect usability, visual comfort, and aesthetic preference (Vinciūnas & Kapočius, 2023). Some studies suggest that light mode may enhance readability due to higher perceived contrast, while others report that dark mode can reduce eye strain and improve visual comfort in certain contexts (Kim et al., 2019). However, much of this research has focused either on subjective user perceptions or objective performance outcomes rather than investigating both simultaneously (Vinciūnas & Kapočius, 2023).

At the same time, cognitive load theory highlights that users' working memory is limited, and that unnecessary design elements may increase extraneous cognitive load, thereby affecting task performance and perceived effort (Sweller, 1988). Yet, there is limited empirical research examining how subjective readability aligns, or conflicts with actual mental effort when interacting with dark and light modes in digital interfaces.

Understanding how dark mode and light mode affect cognitive load can provide designers with knowledge to create effective interfaces that enhance the user experience.

This thesis therefore aims to investigate the relationship between subjective readability and cognitive load in the context of dark mode and light mode. By combining subjective measures of readability with assessments of mental effort, this study aims to provide a more integrated understanding of how interface color themes influence user experience in content-heavy digital environments.

1.1 Problem statement

Digital interfaces are used daily for information consumption, study, and work. A noticeable design choice in modern interfaces is the implementation of dark mode and light mode, which may influence users' subjective readability and cognitive load during extended use. Although both color themes are widely used across devices and applications, previous studies show mixed results regarding which theme is better, especially in content-heavy interfaces. User experience is further affected by factors such as contrast, typography, and cognitive load, which represent distinct but related aspects of interaction.

Cognitive load refers to the mental effort required to process information, and it is constrained by the limited capacity of working memory, which can handle approximately 2-5 units of information at a time (Orru & Longo, 2019). Cognitive load is commonly divided into intrinsic load, determined by the complexity of the content, and extraneous load, caused by

the interface design or other irrelevant factors that do not contribute to learning or task performance. Extreme extraneous load can impact comprehension and reduce efficiency, highlighting the importance of the interface design choices such as color themes.

Moreover, previous research suggests that light mode often results in lower cognitive load and faster task performance than dark mode. However, most studies focus primarily on objective performance measures rather than users' subjective experiences. Subjective readability, defined as users' perceived ease or difficulty of reading text on a display, is an underexplored aspect of interface evaluation. While text-background color combinations affect subjective readability and aesthetics opinions, few studies have examined how dark and light modes influence subjective readability in relation to cognitive load (Gazit et al., 2025).

Although there is a significant amount of research done about dark and light mode on objective performance and cognitive load as well as subjective preferences. There is a significant gap in the research and few studies measure both cognitive load and subjective readability simultaneously and their relationship in the context of dark and light mode. Previous research highlights the need for more studies about the relation between objective workload measurements and subjective questionnaires (Ettling et al. 2025).

1.2 Purpose and research questions

This chapter presents the purpose of this study and the research questions that lay the foundation for it. The study aims to examine how dark mode and light mode influence subjective readability and cognitive load, with a particular focus on extraneous load, and how these variables relate to each other in digital interfaces.

Drawing on the problem statement, previous research suggests that although dark and light mode in digital interfaces have been widely studied, the majority of the research has focused either on objective performance or subjective perception. Limited research measures both cognitive load and subjective readability simultaneously and their relationship in the context of dark and light mode. It is further evident that there is a gap in research when it comes to the relationship between subjective readability and cognitive load in the context of dark mode and light mode. Consequently, the purpose of this study is:

> To investigate and compare how dark mode and light mode in digital interfaces affect users’ subjective readability and cognitive load as the primary variables, while also considering their contribution to overall user experience. Furthermore, the study aims to examine the relationship between subjective readability and cognitive load by focusing on subjective understanding and cognitive demands.

In order to fulfill this purpose, two research questions have been defined

Given that previous findings regarding the effects of dark mode and light mode are inconsistent and influenced by numerous factors, it is necessary to examine these interface themes through two central aspects of digital reading tasks: subjective readability and cognitive load. Hence, the study's first research question is:

[1] How do dark and light interface themes influence subjective readability and cognitive load in content-heavy digital interfaces?

During preliminary research and literature reviews, it became evident that there is insufficient research done about dark mode and light mode in contexts where subjective readability and cognitive load are measured simultaneously. Cognitive load theory explains that increased extraneous load challenges users comprehension while subjective readability reflects users perceived clarity and ease of processing text. Despite the connections between the two factors, few studies have researched how they correlate to each other in the context of dark mode and light mode in digital interfaces. Hence, the study's second research question is:

[2] How are these factors related during extended digital reading tasks?

Together, these research questions allow the study to fulfil its purpose together with qualitative and quantitative research methods. This provides a comprehensive understanding of how dark mode and light mode influence subjective readability, cognitive load and user experience in content-heavy digital interfaces.

1.3 Scope and delimitations

This study focuses on how dark mode and light mode in digital interfaces affects users' subjective readability and cognitive load in a content-heavy environment. The research is limited to a one-page website designed to simulate a realistic fashion blog in order to provide meaningful reading and visual conditions. Participants engage in tasks that enforce information retrieval by using the working memory and continuous reading. Rather than examining objective readability performance, this study examines users perceived readability in correlation with overall cognitive load.

The research was conducted using an experimental within-subject design with a convenience sampling method. The participants consisted of university students at Jönköping University in ages 19 to 30 to gain experiences from young and a digitally knowledgeable population. The interfaces used in the study were two prototypes developed in Figma with the same contents and variables except different color themes.

Several delimitations were made throughout the study to maintain cohesion and feasibility. The study did not include participants with visual impairments or lack of understanding the language of the reading material in order to avoid unwanted influence in the results. In addition, the experiments were conducted individually in a private, controlled indoor setting with consistent lighting conditions in order to minimize variations in environments and ambient lighting that could influence participants in unnecessary ways. This means that the results may not fully reflect the diverse real world usage scenarios of digital interfaces where lighting conditions vary. The research, as stated previously, focuses on subjective readability meaning that it relies on subjective self-reported measures of readability and cognitive load rather than performance based measurements. While this method provides first-hand insight into users own perceived experiences, the results may not fully indicate actual cognitive processing or reading ability.

1.4 Disposition

The remainder of this thesis is structured as follows. Chapter Two explores the theoretical background of this study followed by Chapter Three which outlines the chosen methodology including the methods of data collection and analysis, considerations of validity and reliability. Subsequently, Chapter Four presents the results of the study followed by a comprehensive discussion in Chapter Five. Chapter Six is dedicated to the conclusion of the thesis success by references and appendices in Chapter Seven to Eight.

2 Theoretical framework

This chapter presents the theoretical framework related to the study, including readability in digital interfaces, subjective readability, Cognitive Load Theory, user experience and user control.

2.1 Dark Mode and Light Mode in Digital Interfaces

Dark mode and light mode are two visual presentation styles in digital interfaces, primarily differing in background and text color configurations. Light mode conventionally presents dark text on a light background, whereas dark mode reverses this relationship by displaying light text on a dark background. These modes are extensively implemented across operating systems, web platforms, and software applications, often justified by considerations related to readability, visual comfort, energy efficiency, and user preferences.

Beyond their technical differences, dark and light mode represents several visual environments that may impact how users perceive, process and interact with digital content. The choice of color scheme can shape interface aesthetics, clarity and the subjective experience of working within a digital system. In recent years, dark mode has increasingly been offered as an alternative to the traditionally dominant light interface (Koning & Junger, 2021). Yet, despite their widespread adoption, it remains unclear whether one mode can be considered objectively dominant in terms of usability and user experience across diverse work scenarios, tasks, and demographics.

Previous research has investigated the effects of dark and light modes on readability, visual comfort and user experience. Some studies indicate that light mode may support better readability, particularly during text-intensive tasks, due to higher perceived contrasts. Conversely, other research suggests that dark mode can reduce eye strain and enhance comfort, especially in low-light conditions (Atsani, Mukaromah & Anugerah, 2025). However, findings remain inconsistent, and prior research has mainly focused on isolated effects rather than combined psychological and performance outcomes. Relatively few studies have explored interfaces that allow users to adjust the intensity of dark and light modes, leaving a limited understanding of how user-controlled intensity influences readability, cognitive load, and overall user experience (Atsani, Mukaromah & Anugerah, 2025).

2.2 Readability in Digital Interfaces

Readability is a central concept in the design of digital interfaces and refers to the ease with which written text can be read and understood. In digital contexts, readability is affected by various visual and structural factors, including typography, font size, line spacing, layout and the contrast between text and background. Because digital text is presented on illuminated screens and used under varying lighting conditions, perceptual factors play a key role in shaping how information is processed.

Readability in digital environments is therefore not determined by a single design element, but rather by the interaction between multiple visual and structural variables. Layout configuration, line length and spacing influence eye movement patterns and reading flow,

while typographic characteristics affect the perceptual clarity of letterforms. When these elements are poorly configured, reading may require greater visual efforts, potentially reducing both efficiency and comprehension. Consequently, readability can be understood as a perceptual and cognitive outcome shaped by interface design decisions.

Moreover, luminance contrast between text and background has consistently been identified as a key predictor of reading performance. Higher contrast levels have generally been associated with improved reading speed, accuracy, and perceived ease of reading. However, contrast alone does not fully explain readability outcomes, as the polarity of the color configuration, whether the text is presented as dark on light or light on dark, also appears to influence visual processing (Hall & Hanna, 2007).

Studies exploring visual polarity suggest that positive polarity (dark text on a light background) often results in better reading performance compared to negative polarity (light text on a dark background). These findings indicate that text-background configuration is not an aesthetic preference, but a design factor with measurable perceptual implications (Buchner & Baumgartner, 2007).

Overall, these theoretical insights highlight the importance of contrast and polarity when evaluating readability in digital interfaces. This makes the comparison between dark and light interface modes particularly relevant when investigating how users interact with text-based content.

2.3 Subjective readability

Subjective readability can be understood as users perceived ease of reading and understanding visual content (Cabouat et al., 2025). Unlike objective readability measures (e.g., reading speed or accuracy), subjective readability focuses on the user's subjective experience, which is formed by both the visual design and individual cognitive processes.

Research within visualization and human-computer interaction highlights that readability is not just a property of the interface itself, but rather appears from the interaction between the visual display, the individual user, and the task context. According to Cabouat et al. in *PREVis: Perceived Readability Evaluation for Visualizations*, subjective readability can be understood as a multidimensional construct that includes several key components.

The PREVis framework identifies four main dimensions:

- Understandability: how easily users can interpret the content and understand how information is presented
- Layout clarity: how visually organized, non-crowded, and non-dristracting the interface appears
- Data readability: how easily users can locate and retrieve information
- Feature readability: how clearly patterns, trends, or important elements are perceived

These dimensions highlight that readability is closely linked to both visual structure and cognitive effort. A design that is messy, poorly contrasted, or visually overwhelming can

increase extraneous cognitive load and reduce subjective readability (Sweller, Ayres & Kalyuga, 2011).

This is particularly relevant when comparing light mode and dark mode interfaces. Visual characters such as contrast, brightness, and color schemes differ between these modes and can influence how text and content are perceived.

From a cognitive perspective, subjective readability is also connected to cognitive load theory, which suggests that users have limited mental resources when processing information. Interfaces that are easier to scan, visually clear, and well-structured reduce unnecessary cognitive effort, allowing users to focus on understanding the content (Paas, F., van Gog, T. & Sweller, J. 2010).

2.4 Cognitive Load Theory

Cognitive load theory (CLT) is a theory based on the concept of cognitive load. The concept of cognitive load refers to the amount of mental resources used by users when processing information. The basis of the theory is that the human mind has limits when processing information. Different users can have varying kinds of mental resources that are used when processing information and interface designs. When an interface is complex or needs more mental resources from users to understand, the performance lowers and the brain gets exhausted (Whitenton, K. 2014).

This is why the design of interfaces are important. To create interfaces that are usable and accessible to users, designers need to understand how cognitive load affects users. CLT provides a framework for understanding how design can affect the human memory by overwhelming the working memory. It focuses on how to control the overload of information to make sure users can process it efficiently by providing strategies that manage cognitive load. CLT is grounded in human cognitive architecture which consists of the long term memory and the working memory, memory systems that handles the cognitive load. The long term memory is unlimited and stores information that you acquire through a daily basis. It's viewed as the central structure of human cognition and contains the knowledge humans gather. The working memory processes the information that's stored or is on its way to being stored in the long term memory (Paas, F., van Gog, T. & Sweller, J. 2010).

CLT identifies three categories of cognitive load: intrinsic, extraneous and germane. Intrinsic load refers to the cognitive effort required by the complexity of the material itself. It depends on the number of elements that are processed at the same time and how they interact such as reading content-heavy webpages. Extraneous load refers to the cognitive load caused by poor design that challenges comprehension. It's due to the design of the environment or interface and not the material itself such as the bad contrast ratio. Germane load refers to the working memory resources to understand and store intrinsic load. It's about processing and understanding information in long term memory such as trying to understand and remember concepts from a webpage. When an interface is easy to navigate, the user can minimize extraneous load and focus mental resources on germane load and remembering the content (Paas, F., van Gog, T. & Sweller, J. 2010).

Intrinsic and extraneous cognitive loads are additive. They work together to determine the total cognitive load imposed by the material a user interacts with. Both types draw on the same working memory resources, although effort is divided between intrinsic load (supported by germane resources) and extraneous load (interacting with its presentation), they come from the same working memory pool. If the total cognitive load exceeds the available capacity of the working memory, the cognitive system becomes overloaded and will fail to process the information. In cases of poor instructional design that produces high extraneous load, an overwhelming amount of capacity may be consumed by coping with the presentation and consequently insufficient resources remain to process the intrinsic load. As a result, users fail to understand the content (Sweller, Ayres & Kalyuga, 2011).

Instructional design should aim to reduce extraneous cognitive load in order to reduce working memory resources spent on unnecessary processing, leaving more capacity available for germane resources to handle intrinsic cognitive load. The goal is to make sure that the majority of working memory can be devoted to learning the content rather than getting overwhelmed by poor presentation or design (Sweller, Ayres & Kalyuga, 2011).

2.5 User Experience and User Control

User experience (UX) refers to the overall experience a person has when interacting with a digital system, perceptions of usability, satisfaction, efficiency and enjoyment (Norman, 2013). An important component of UX is the sense of control that users have over the interface and their interactions with it. When users can adjust settings, navigate and modify the system to their preferences, they are more likely to feel comfortable, competent, and engaged. In contrast, a lack of control or confusing interface elements can increase cognitive effort, reduce satisfaction and negatively impact task performance.

Even though it's fairly agreed that experience as a concept is personal, individuals perceive and feel experiences differently depending on their context, when it comes to the concept of UX, Quaresma, Soares, and Correia (2022) identifies four main perspectives. The first perspective views experience as a result of the user's perception when interacting with digital softwares of interfaces. The second perspective expands this view and defines the user's experience as the result of the user's perception when interacting with any product or service, it's not limited to digital services. The third perspective is that user experience is a method in a design process and lastly the fourth perspective conceptualizes the users experience with a broader system not just a product (Quaresma, Soares & Correia, 2022).

2.6 Summary of the theoretical framework

Dark mode and light mode are widely used color styles in digital interfaces today, and even though it's a widely researched subject, there are a variety of different findings regarding their effects. They can affect readability, cognitive load and user experience differently (Atsani, Mukaromah & Anugerah, 2025).

When it comes to readability in digital interfaces, it's affected by different factors such as contrast, layout and typography (Hall & Hanna, 2007. Readability can be seen as both objective and subjective, where subjective readability explains how easy a user feels the content is to understand and read.

Cognitive Load Theory explains how interfaces and their design can affect users working memory and its limited resources, especially through extraneous load caused by poor design that hinders users comprehension (Paas, F., van Gog, T. & Sweller, J. 2010).

User experience and user control can further affect how a user interprets and interacts with digital interfaces. Higher user control over the interface can improve how comfortable the user feels with the system, how satisfied they are with it while also improving the usability (Quaresma, Soares & Correia, 2022).

2.7 Conceptual definitions

To ensure clarity and coherence throughout the study, the key concepts are defined as followed:

Subjective readability

Is a primary variable in this study and refers to the users perceived ease of reading and understanding content in digital interfaces. In this study, subjective readability is measured through users' self-reported survey ratings related to clarity of text and visual elements, ease of scanning, text comfort and overall readability.

Cognitive load

Is a primary variable in this study and refers to the mental effort required to process information. In this study, this is measured using self-reported mental effort ratings collected via survey questions inspired by NASA-TLX.

Extraneous cognitive load

Refers to the part of cognitive load caused by the design of the interface, rather than the content itself, that can hinder users comprehension. In this study, extraneous load is used as a subcategory of cognitive load and is not measured directly. It's used as a theoretical concept to interpret differences in overall cognitive load between dark mode and light mode interfaces.

Visual comfort

Refers to how comfortable a digital interface feels for the users eyes when reading or interacting with content. It is influenced by factors such as contrast, brightness, and color combinations. In this study, visual comfort is evaluated through users self-reported perceptions and is treated as a contributing factor to overall user experience.

User experience

Refers to the overall experience a user has when interacting with a digital system, including insights of usability, efficiency, satisfaction, and emotional response. In this study, user experience is evaluated through subjective ratings and qualitative feedback related to participants' interactions with the interface designs.

Usability

Refers to how effectively, efficiently, and satisfactorily a user can achieve specific goals within a system. In this study, usability is reflected through the participants ability to complete tasks, as well as their perceived ease of use and interaction with the interface.

To clarify the relationship between the concepts, subjective readability and cognitive load are treated as the primary variables of the study. Visual comfort and usability are considered related but secondary factors that may influence these variables. User experience is used as a general concept that captures the participants overall perception of the interface. This ensures that each concept is analytically separated while still acknowledging their interdependence.

3 Method and implementation

This chapter describes and motivates the methodological approach used in this study. It explains the study design, the reasoning behind the chosen methods, how data were collected and analyzed, and how validity and reliability were ensured. The study uses a mixed-method approach to explore how dark mode and light mode affect subjective readability and cognitive load in content-heavy digital interfaces.

3.1 Research design

An experimental within-subjects design was chosen for this study. Each participant interacted with two digital interface prototypes: one in dark mode (light text on dark background) and one in light mode (dark text on light background). This design allows each participant to have their own control, reducing the influence of individual differences on subjective readability and cognitive load.

The study combined quantitative and qualitative methods. Quantitative measures captured subjective ratings of readability and cognitive load using Likert-scale items, while qualitative data were collected through open-ended questions to provide deeper insights into participants' experiences and preferences.

The within subject-design also allowed the research to control for confounding factors such as prior digital reading experience or personal interface preferences. To improve order effects, the outcome of prototype exposure was counterbalanced across participants (Creswell, 2014).

3.2 Participants

The participants were recruited among students registered at Jönköping University. A total of 22 students participated in the study, aged between 19 and 30, representing various academic programs.

The choice of university students as the study population was intentional. Students frequently engage with content-heavy digital materials such as academic articles, online learning platforms, and digital course literature. Therefore, they represent a relevant user group for investigating subjective readability and cognitive load in digital reading environments.

Additional criteria required the participants to have normal vision and sufficient proficiency in the language of the reading material. The participants with known color vision deficiencies were excluded to avoid confounding effects related to color perception.

The sampling method can be described as a convenience sample, which is common in experimental research within educational settings. While this approach supports feasibility and controlled data collection, it may limit the generalizability of the findings to broader populations beyond university students. This limitation is further discussed in the section on validity and reliability (Ahmed, 2024).

3.3 Materials

Two prototypes of a fashion blog website interface were developed using Figma for this study due to its text heavy nature, identical in content, layout, typography, and spacing, differing only in color scheme:

- Dark mode: light text on dark background
- Light mode: dark text on light background

The text content was designed to simulate realistic reading conditions in content-heavy digital environments. The text included information about fashion, articles and news about the latest collections combined with pictures to imitate a realistic fashion blog. Fonts, line spacing, and layout were structured to minimize variation in extraneous cognitive load. The text length is approximately 500 words per page, divided into paragraphs and titles in order to encourage continuous reading by participants while minimizing fatigue.

To estimate subjective readability and cognitive load, a survey was provided after each prototype interaction, consisting of:

- Likert-scale items evaluating subjective readability, clarity, and mental effort
- Simplified NASA-TLX items measuring cognitive load

- Open-ended questions for qualitative feedback on user experience and preferences

All responses were collected anonymously using online forms interacted with the prototypes.

3.4 Prototype design

In order to ensure accessibility, the prototype of the website was designed in accordance with the W3C guidelines and recommendations for web accessibility. Guideline G18 recommends a minimum contrast ratio of at least 4.5:1between text and background in order to ensure that users are able to read the text that is presented over a background in the prototype. This requirement was implemented and verified using a Figma Plugin (Stark - Contrast & Accessibility checker) to guarantee correct contrast ratios throughout the prototype (W3C, n.d.).

Furthermore, throughout the study, typography choices were carefully thought out and made with the purpose to enhance readability and accessibility in accordance with Success Criterion 1.4.3 (Contrast (Minimum)) and Success Criterion 1.4.12 (Text spacing). A base font size of 20px was used for body text with a line height of 28px, while a font size of 16px was applied to smaller links with a line height of 24px. Headings ranged from 32px to 90px to establish a clear visual hierarchy (W3C, n.d.).

Since this study is based around dark mode and light mode in digital interfaces, the choice of color played an important part. Previous research and guidelines show that using pure black (#000000) and pure white (#FFFFFF) is not recommended in UI design. This high contrast can result in halation for users with astigmatism when white text is on a pure black background and make it hard to read (Sharma, 2026). Using pure white and black together exceeds the accessibility minimum of 4.5:1 that is recommended by the W3C, (W3C, n.d.) as well as causing eye strain due to the extreme luminance difference. It is due to this that it is recommended to use dark gray for backgrounds and off black for text with white backgrounds in order to reduce the harsh and high contrast (Humbert, 20206). Apple's dark mode and light mode remain in line with accessible contrast ratios. By using off white background for light mode and dark gray for dark mode in their UI, they ensure good contrast between elements and reduced eye strain for users (Apple, n.d.).

The colors that were chosen for the two prototypes are based on the Atlassians design system guidelines on choosing colors for light and dark themes. In UI design, light and dark color themes and modes use different color shades in order for the elements to remain readable and clear. The Atlassian color palette provides two neutral ramps, one for light mode and one for dark mode. These ramps are designed to ensure easy conversion between color themes, each color shade in the light mode ramp has a matching shade in the dark mode ramp, and vice versa. Neutral100 in the light mode palette equates to DarkNeutral100 in the dark neutral palette. These two shades were used in the prototype design with the purpose of ensuring consistent perceived visual contrast when switching between modes while also aligning with Success Criterion 1.4.3 (Contrast (Minimum)) from the W3C guidelines (Atlassian, n.d).

For the dark mode version, #1F1F21(DarkNeutral100) was used for the background of the webpage while #F8F8F8 (Neutral100) was used for the font color of all text on the page (see figure 1). For the light mode version, #F8F8F8 (Neutral100) was used for the background and #1F1F21(DarkNeutral100) was used for the font color of all the text on the webpage (see figure 1)(Atlassian, n.d).



Figure 1: Primary color used in the prototypes

3.5 Survey design

In order to further investigate participants' preferences and experiences with the two different prototypes, a survey was created as a means for the participants to further motivate and reflect on their opinions. The survey was used as an additional tool for the study to gain insight into users preferences of dark mode and light mode through their own words and thoughts besides the usability tests and observations.

The survey was designed in parts where the first part focused on task-based questions, the second part on readability, third part on cognitive load, fourth part on visual comfort and the last part on overall user experience.

The second part of the survey was built upon the Web Accessibility guidelines WCAG guidelines, more specifically Success Criterion 1.4.3 (Contrast (Minimum)) and Success Criterion 1.4.12 (Text Spacing). These guidelines define the importance of readable text through sufficient contrast and text spacing and were adapted into the user-centered survey questions to capture perceived readability by asking users statements related to text readability, contrast and text design. The survey included statements such as “The text size was comfortable to read”, “The spacing between lines and paragraphs improved readability” and “The text on the page was easy to read” to reflect these criteria.

In addition, the survey's third part focused on cognitive load and was inspired by the NASA task load index, a tool to measure perceived workload using multiple rating scales. However, rather than adopting its original scale rating technique, the tool was primarily used to design relevant survey questions (NASA Human Systems Integration Division, n.d.).

The following questions in the survey were inspired by this framework: “How mentally demanding were the tasks?”, “How successful were you in accomplishing the tasks?”, “I could read without feeling mentally tired”, “It was easy to understand the content”, and “ How overwhelmed were you when you did the tasks?” are the questions inspired by the NASA task load index (NASA Human Systems Integration Division, n.d.).

The purpose of this section in the survey was to measure the subjective mental workload experienced by users while conducting the prototype tasks (NASA Human Systems Integration Division, n.d.).

In order to measure participants' perceived mental effort, the rating scale technique used in the cognitive load part of the survey was based on the mental effort scale developed by Fred G. W. Pass, (1994) for measuring perceived task difficulty. This scale is a modified version of an earlier rating scale developed by Bratfisch, Borg, and Dornics scale (1972) for measuring perceived task difficulty. The scale ranges from 1 to 9, and the participants reported the amount of mental effort they experienced by selecting a value on a 9-point category scale. This allowed them to translate their perceived mental effort amount into a numerical label with assigned categories running from ”1” that corresponds to “very, very low mental effort” to “9” that corresponds to “very, very high mental effort”. (Paas, van Merriënboer, & Adam, 1994)

Due to the need for coherency throughout the survey, all scale questions were adapted to use this unified rating scale technique with the 9-point category, with minor modifications made on the corresponding category labels in order to maintain relevance to the question context.

3.6 Procedure

To ensure controlled conditions for the experiment and gather fair and reliable results, the experiments were conducted individually with one participant and one observer present in a silent, distraction-free indoor setting. The experiments took place in a standard room at Jönköping University with consistent overhead lighting and no outdoor light sources in order to minimize variations in ambient lighting that could influence participants cognitive load and subjective readability.

The experiment was conducted as follows:

1. The participants were introduced to the study, its purpose, the procedure, and informed permission was acquired.
2. The participants interacted with the first prototype (dark or light mode) and completed a reading task of approximately 10-15 minutes.
3. After completing the task, the participants complemented the corresponding survey individually with no observer present in order to not be affected by outside influence.
4. A short break was provided before the participants interacted with the second prototype and completed the second survey.
5. At the end, the participants submitted demographic information and general reading habits.

The order of the prototype presentation was neutralized to reduce potential order effects. Half of the participants started with the light mode interface, while the other half began with the dark mode interface.

3.7 Task questions

In this chapter, the tasks that participants performed out loud in the beginning, upon first seeing each of the prototypes are presented:

- What type of website do you think this is?
- What is the main goal of the page?
- Which brand was mentioned in the article?
- Were you able to find the link to the contact page?

3.8 Survey questions

In this chapter, the survey questions used in this research are presented:

General questions at the beginning

- Do you agree to participate in this study and allow your responses to be used for academic research?
- Which version of the prototype did you use?

For subjective readability:

- The text on the page was easy to read
- I could easily scan and find information
- The text size was comfortable to read
- The spacing between lines and paragraphs improved readability
- The contrast between text and background made the text easy to read

For cognitive load

- How mentally demanding were the tasks?
- It was easy to understand the content
- I could read without feeling mentally tired
- How successful were you in accomplishing the tasks?
- How overwhelmed were you when you did the tasks?

For visual comfort

- The design felt comfortable for my eyes
- I could read for a longer time without discomfort

Overall UX

- The web page was pleasant to read
- I would prefer reading articles in this visual style (with these colors)

After both of the prototypes to compare

- Which version did you prefer?
- Which version made the text easier to read?
- Which version felt less mentally demanding?
- Which design did you find more visually appealing?

Open-ended

- What did you like most about the web page you just used?
- What did you like least?
- Did the color theme affect your reading experience? How?

3.9 Data collection

The survey was structured with the purpose of gathering data aimed at addressing the study's first research question. It consisted of 26 questions in total divided into six sections. The first section included four task related questions. The second section consisted of five subjective readability related questions. The third section comprised five questions focused on cognitive load related. The fourth section consisted of two questions addressing visual comfort. The fifth section consisted of five overall user experience related questions. Finally, the sixth section included four questions comparing dark mode and light mode.

Of the 26 questions, 14 were Likert-scale questions (e.g., 1-9) to enable quantitative analysis while six were open ended questions to capture qualitative insights. The remaining six questions consisted of binary choices.

The quantitative data were collected to answer the research questions regarding subjective readability and cognitive load. Likert-scale items provided numerical measures of participants' perceived ease of reading and mental effort in each interface condition. This approach enabled direct comparison between dark mode and light mode conditions.

The qualitative data were collected through open-ended questions, allowing the participants to describe their experiences, highlight difficulties, and express preferences. These responses were used to compliment quantitative findings and provide a deeper understanding of the participants' subjective experiences.

The data collection process was documented to ensure that all participants experienced the same conditions. Online surveys were used to maintain anonymity and ensure secure data storage.

Each research question was connected to a specific data collection instrument:

- RQ 1 (subjective readability and cognitive load): Likert-scale readability items and NASA-TLX items measuring perceived mental effort.
- RQ 2 (relationship between readability and cognitive load): Combined analysis of quantitative ratings and qualitative feedback.

3.10 Data analysis

This chapter presents the methods used to conduct the analysis of the collected data in the study. Based on the methods used for collecting data, appropriate analysis methods were chosen to get meaningful results and interpret participant responses.

Quantitative data about subjective readability and cognitive load were gathered from the survey consisting of Likert-scale ratings that measured users opinions and experiences on each prototype. Readability related items and cognitive load items, inspired by the NASA-TLX were aggregated into composite scores. These results are shown in figure 2 and 3.

Furthermore, descriptive statistics were then calculated from the Likert-scale responses by calculating the mean scores to summarize the results for each interface condition.

T-tests were carried out to analyze the quantitative data further. However, it is notable that one participant was excluded in the t-tests due to incomplete paired responses. The analysis of the results in this study was limited to descriptive statistics from the Likert-scale responses as well as paired samples t-test and correlation analysis. No additional inferential statistical tests were used.

Figure 3 presents the composite scores for subjective readability and cognitive load in dark mode for each participant. Each value represents the mean score calculated from the corresponding subjective readability and cognitive load Likert-scale questions.

In order to get consistency in measurement of cognitive load , negative items were reverse coded prior to analysis, this results in higher scores indicating lower perceived cognitive load. Response patterns were analyzed and put into visual graphs to better understand the data.

Due to the study using a within subject design, paired comparisons were conducted to show the differences between dark and light mode. This method was used to compare and rank users preferences and mimics users decision making in real life by having them compare options based on personal references.

In order to analyse the relationship between subjective readability and cognitive load, a correlation analysis was conducted to determine if a relationship between the two factors exists and if so, how strong it is. This method was used to analyse quantitative data collected from the experiment and showed whether higher subjective readability was connected with lower mental effort or vice versa.

Although cognitive load is measured as an overall concept, differences between dark and light modes are interpreted as primarily reflecting variations in extraneous cognitive load. This assumption is based on the fact that the content, structure, and task complexity of the prototypes were kept identical, while only the visual design (color theme) differed.

The qualitative response from open-ended questions were analyzed using thematic analysis. The responses were gathered and categorized into themes reflecting patterns in experiences, opinions and perceptions.

The qualitative responses from the open-ended questions were analyzed by grouping the participants' answers into themes, following a thematic analysis. The analysis focused on identifying patterns related to the participants' understanding of the website type, perceived purpose of the page, brand recognition, and navigation experience. The responses were reviewed and categorized based on similarities in descriptions, opinions, and interpretations in order to identify common experiences across the participants.

Due to the mixed-method design, quantitative and qualitative findings were analyzed together to provide a more comprehensive understanding of the participants' experiences. Numerical results showed measurable comparisons between conditions, while qualitative feedback provided an understanding of participants' subjective experiences.

Since this study used a mixed-method approach, the quantitative and qualitative findings were analyzed together to provide a broader understanding of the participants' experiences. The quantitative data provided measurable comparisons between dark mode and light mode regarding subjective readability and cognitive load, while the qualitative responses contributed deeper insights into the participants perceptions, preferences, and interpretations of the interfaces. Combining these methods allowed the findings to be interpreted from both statistical and experimental perspectives.

3.11 Validity and reliability

The validity of this study concerns the extent to which methods used actually measure what they are intended to measure, specifically subjective readability and cognitive load in relation to dark and light mode interfaces. To strengthen internal validity, the study applied a within-subject design where each participant interacted with both interface conditions. This reduces the influence of individual differences, as each participant performs as their own control. Additionally, the prototypes were identical in all aspects except the color theme, which helps isolate the effect of dark and light mode.

To further support construct validity, established measurement approaches were used. Subjective readability was measured through multiple Likert-scale items based on recognized usability and readability principles, while cognitive load was evaluated using items inspired by the NASA-TLX and mental effort scales. However, since cognitive load was measured through self-reported data, there is a risk that the results reflect subjective rather than actual mental effort.

External validity is limited due to the use of a convenience sample consisting of university students. While this group is relevant for content-heavy digital reading tasks, the finding may not fully represent other demographics with different backgrounds, ages, or knowledge of digital experience. The controlled experimental setting could also differ from real-world environments where lighting conditions, device types and distractions vary.

Moreover, to enhance reliability, all participants followed the same procedure under similar conditions, including identical instructions, controlled lighting, standardized screen brightness levels and equal exposure time to both prototypes. The use of standardized survey questions

and consistent rating scales further supports reliable data collection. However, subjective measures may still introduce variability, as the participants may interpret questions differently or be influenced by temporary factors such as fatigue or mood.

Overall, while the study demonstrates logical levels of validity and reliability, the findings should be interpreted with consideration of the limitations related to sample selection, self-reported data, and experimental conditions.

3.12 Considerations

Throughout this research, ethical considerations and principles were prioritised. According to the Swedish Research Councils (2025) guidelines for good research practice, the participants' protection is a priority in research. Following this, the experiments and survey avoided fatigue, eye strain and stress as much as possible in order to prioritise the participants' integrity. The SRC says that irrespective of what type of research is done, the protection of the participants shall always be prioritised (Swedish Research Council, 2025).

As the study involved screen-based reading tasks, particular consideration was given to the participants' physical comfort in order to follow the SRCs principles. The duration of each session was kept reasonable, approximately 10-15 minutes per prototype, in order to minimise unnecessary fatigue. A short break was also provided between the two prototype conditions. In addition, the experiment was conducted in a calm indoor environment with controlled lighting conditions to reduce discomfort and avoid external distractions that could affect the participants.

As well as ensuring the protection of the participants, the Swedish Research Council lays a basis for ethical considerations for human involvement in research. This is based on four fundamental principles: To do good (beneficence), To do no harm (non-maleficence), To respect autonomy and To uphold Justice (Swedish Research Council, 2025).

In line with these principles, all participants were informed about the purpose of the study, the procedure of the experiment, and the estimated time required before participation began. The participation was entirely voluntary, and informed consent was obtained from all participants before the data collection. The participants were also informed that they had the right to withdraw from the study any time without explanation or consequences.

In research, an essential ethical principle is the requirement to obtain informed consent from participants. This requires that the participants are provided with clear and sufficient information about the purpose and procedure of the study, their right to withdraw at any time, potential risks and how their data will be handled in the study(Swedish Research Council, 2025).

The General Data Protection Regulation (GDPR) is a privacy and security law within the European Union that governs the collection, usage and storage of personal data. It established key regulatory principles for data protection that includes lawfulness, accuracy and accountability (GDPR.eu, n.d).

In this study, data collection was conducted through digital surveys and handled in accordance with the General Data Protection Regulation principles. The participants remained anonymous, and no personally identifiable information was collected during the study.

The collected data were used only for academic purposes related to this thesis and were only accessible to the authors. The results are presented on a general level, ensuring that no individual participant can be identified. Furthermore, the survey began with an informed consent question, which provided participants with information about their rights to follow the Swedish Research Councils and General Data Protection principles.

The Swedish Research council (2025) highlights the identification of protection interests and potential risks as an important part of ethical research. Researchers are responsible for safeguarding the health, safety and well-being of all individuals affected by the research as well as identify and manage potential risks and minimise negative consequences. These principles guided the design decisions of this study. Potential risks such as fatigue, eye strain and cognitive load were identified as participants interacted with digital interfaces and surveys.

The prototypes were designed according to established accessibility guidelines by the WCAG, including appropriate contrast ratio, readable typography, and sufficient spacing between text elements. This was done to reduce the risk of eye strain or cognitive load caused by bad interface design as well as avoiding further harmful stimuli such as uncomfortable contrast for the users. Further design choices for the prototypes were carefully considered. Given the color selection, no pure white and pure black was used to avoid extreme brightness and visual discomfort.

Finally, all participants were treated equally throughout the experiment. They received the same instructions, completed the same tasks, and interacted with prototypes that only differed in color theme. This ensured fair procedure while also strengthening the ethical responsibility of the study in a respectful way.

4 Results

This part presents the results of the study based on both quantitative and qualitative data. The quantitative results are based on Likert-scale responses measuring subjective readability, cognitive load, and user experience, while qualitative insights are based on participants' open-ended responses.

The results are structured to address the study's research questions: “How do dark and light interface themes influence subjective readability and cognitive load in content-heavy digital interfaces?” and “ How are these factors related during extended digital reading tasks?” .

This chapter will present the data collected through a survey and experimental procedure conducted in this study. A total of 22 participants were involved in this study consisting of students from Jönköping university between the ages of 19-30.

4.1 Analysis of Likert-scale questions

For light mode, subjective readability scores ranged from 4.8 to 8.8 with a mean of 7.40 (SD = 1.19). For dark mode, subjective readability scores ranged from 6.0 to 9.0, with a mean of 7.78 (SD = 0.92).

For cognitive load, the scores in light mode ranged from 5 to 8.6 with a mean of 6.70 (SD = 1.25), while scores in dark mode ranged from 4.4 to 9, with a mean of 6.80 (SD = 1.47). As negative items were reverse coded prior to analysis, higher scores represent lower perceived cognitive load.

Subjective readability	Mean	SD
Light mode	7.40	1.19
Dark mode	7.78	0.92

Figure 2: Mean and Standard deviation of subjective readability in light mode and dark mode

Cognitive load	Mean	SD
Light mode	6.70	1.25
Dark mode	6.80	1.47

Figure 3: Mean and Standard deviation of cognitive load in light mode and dark mode

4.2 Paired comparisons (t-test)

The results gathered from the experiment and survey reveal high subjective readability in both light mode and dark mode. Dark mode indicated a slightly higher mean score of 7.78 (SD = 0.92) compared to light mode that had a mean score of 7.40 (SD = 1.19).

The lower standard deviation in dark mode exhibits a slight more consistency in responses among participants, whereas the higher standard deviation in light mode displays more variability in results from participants.

Regarding cognitive load, the results reveal minor differences between modes. Based on the participants' responses, the perceived cognitive load is very similar in both modes with slightly higher scores in dark mode indicating lower cognitive load, since cognitive load items were reverse coded prior to analysis, higher scores reflect lower perceived cognitive load. As shown in figure 2 and 3 , dark mode has a mean score of 6.80 (SD = 1.47), in comparison to

light mode that has a mean score of 6.70 (SD = 1.25). The higher standard deviation score in dark mode indicates more variability in responses from the participants, whereas the standard deviation score for light mode was slightly lower, indicating more consistent responses.

Variable	t(df)	p-value	Result
Subjective readability	t(20) = 1.14	.265	Not significant
Cognitive load	t(20) = 0.05	.961	Not significant

Figure 4: Results of t-tests

The results generated from the t-test of the total subjective readability likert-scale questions with both light mode and dark mode are as seen in figure 4. The results did not reveal statistical significance(p = .265).

Similarly, when it came to the cognitive load items in the likert-scale questions, the t-tests showed no statistically significant difference between light mode and dark mode (p = .961).

Based on the results gathered, participants reported similar performance with both dark mode and light mode, specifically regarding mental cognitive load. While the subjective readability score is slightly higher in dark mode, the differences between the two modes were small.

4.3 Correlation analysis

A Pearson correlation analysis was conducted to examine the relationship between subjective readability and cognitive load in dark mode and light mode.

To further strengthen the correlation analysis, a scatter diagram with a fitted trendline was used in order to visualize the relationship between the two variables.

DARKMODE

8,2	6
8,6	9
8	5,6
7	5,8
8	6,8
8	6
9	8,4
7	5,8
9	8,8
6,4	5,2
8	6
8,2	8,6
8,6	9
9	5,2
7	7,6
7	5,4
7,6	7,2
7,4	6,4
6	4,4
6,2	8
8	6
9	9

Figure 5: Mean of subjective readability and cognitive load in dark mode

The results from the correlation analysis show a moderate positive correlation between subjective readability and cognitive load in dark mode (r = 0.52). Cognitive load items were reverse coded prior to analysis, so higher scores reflect lower perceived cognitive load.

Accordingly, higher readability scores were associated with lower perceived cognitive load. As readability increases, perceived cognitive load is lower, meaning that the participants who find it easier to read the content feel lower levels of mental effort needed to process it.

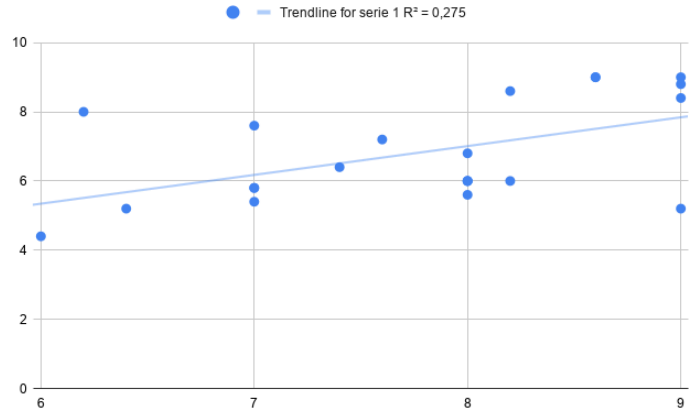


Figure 6: Scatter diagram with a fitted trendline for subjective readability and cognitive load in dark mode

Figure 6 illustrates the relationship between subjective readability and cognitive load in dark mode based on the gathered results using a scatter diagram with a trendline. The distribution of data points shows an upward trend, indicating a positive connection between the two variables.

The coefficient of determination ($R^2 = 0.275$) indicates that approximately 27.5% of the variance in cognitive load is associated with subjective readability. These results are consistent with the scatter diagram and the correlation analysis, which shows a moderate positive correlation between the variables ($r = 0.52$).

LIGHT MODE

5,4	6,4
7,2	8,6
8,4	6,6
7,4	6,2
5	5
8	8,6
9	8,6
7,8	6,8
8,8	8,2
8	6,8
6,6	6,8
7,6	4,6
8,2	7,8
8	6
8,4	6,8
7	7,2
7,8	4,8
4,8	5,8
6,6	6,6
6,8	5,2
8,6	8,2

Figure 7: Mean of subjective readability and cognitive load in light mode

Figure 7 presents the composite scores for subjective readability and cognitive load in light mode for each participant. Each value represents the mean score calculated from the corresponding subjective readability and cognitive load likert-scale questions.

The results gathered from the answers on light mode indicate a moderate positive correlation ($r = 0.49$) between subjective readability and cognitive load. However it's notable that this correlation is slightly weaker than that observed for dark mode ($r = 0.52$), although the difference is small.

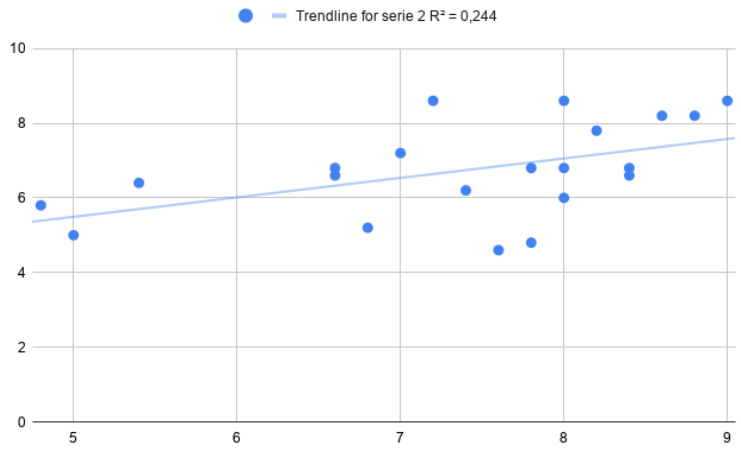


Figure 8: Scatter diagram with a fitted trendline for subjective readability and cognitive load in light mode

Figure 8 illustrates the relationship between subjective readability and cognitive load in light mode based on the gathered results using a scatter diagram with a trendline.

This distribution of data points shows a subtle upward trend, illustrating a positive connection between the two variables. This pattern shows that higher subjective readability scores are connected with lower perceived mental workload. The fitted trendline on the scatter diagram further reinforces this pattern.

The coefficient of determination ($R^2 = 0.244$) indicates that roughly 24.4% of the variance in cognitive load is associated with subjective readability. These results are consistent with the correlation analysis that displays a moderate positive connection between the two variables ($r = 0.49$).

4.4 Cognitive load

The differences between dark mode and light mode are interpreted as variations in extraneous cognitive load, based on cognitive load theory.

Since both versions were identical in terms of content, structure, and task complexity, the only variable that differed between conditions was the visual design, specifically the color theme. Therefore, any differences in user performance or perception can be attributed to the interface design.

However, the results show no clear differences between dark mode and light mode in terms of task performance or user understanding. The participants in both conditions were able to correctly explain the website's purpose, identify its content, and complete the task of locating the contact page.

This indicates that neither interface introduced additional unnecessary mental effort. The high level of task success across both conditions suggests that the interface supported efficient information processing without overloading the users working memory.

Furthermore, minor variation in responses, such as differences in brand recognition or interpretation of the website's purpose, may indicate subtle effects of visual presentation. However, these differences were not consistent enough to suggest a strong impact on cognitive load.

4.5 Qualitative analysis (open-ended responses)

One dominant theme was the understanding of the website type. Most participants consistently described the interface as a fashion-related platform, commonly referring to it as a blog, magazine or inspiration site. This indicates that the overall concept of the interface was clearly communicated.

Table 1 presents the main themes identified from the qualitative analysis of participants' open-ended responses. The table summarizes patterns in user experience, including readability, visual design, usability and brand perception, together with representative participant quotes.

Theme	**Description**	**Summary**	**Quotes from the participants**
Readability and eye comfort	The participants described how easy or difficult the interface was to read and how it affected eye strain.	Mostly positive for both modes, slightly stronger for dark mode comfort.	**Dark:** “Easier for the eyes” “Soft for the eyes” “More calm and pleasant" **Light:** “White background helped me grasp information” “Easier to see the text” “The contrast helped readability”
Dark mode preference	The participants who preferred dark mode often mentioned reduced eye strain and calmness.	Positive	“Dark mode feels more calm” “Less straining for my eyes” “More pleasant for longer reading”
Light mode preference	The participants who preferred light mode described better focus and clarity.	Positive	“Light mode made it easier for me to focus on the text” “Easier to follow the content” “Better clarity for reading”
Layout & structure	Feedback about spacing, sections, and visual organization.	Mostly positive	“I liked that it was divided into sections” “Clear and minimal design”
Brand recognition	Some participants had difficulty identifying or remembering the brand.	Mixed	“I don't know” “Miu Miu?” “Fashion…something”

Visual design	Opinions about overall visual impression and style.	Positive	“Minimal and sleek design” “Visually pleasing” “Modern aesthetics"
Usability/navigation	Comments on how easy it was to use and find information.	Positive	“Easy to use” “Easy to navigate” “Clear UI/UX structure”
Too much text/readability issues	The participants felt some sections were too text-heavy	Negative	“A lot of breadtext” “Too much text” “Long paragraphs made it harder to read”

Table 1: Thematic analysis of the participants' open-ended responses.

The analysis reveals several themes in the participants' responses. A second theme concerned the subjective purpose of the page, where most of the participants described it as providing inspiration and fashion-related content, while a smaller number interpreted it as a more commercial platform.

A third theme was brand recognition, where many participants correctly identified MIUMIU, although some uncertainty was observed.. This argues that although the brand was present, it was not consistently notable for all users.

The final theme related to navigation and usability. Most of the participants reported that they were able to find the contact page, indicating that the navigation structure was generally intuitive. A small number of participants were unable to locate it, suggesting minor usability challenges.

Overall, the qualitative findings show consistent patterns across the participants, with only limited variation in interpretation and recognition.

4.6 Mixed methods

The quantitative results show a high level of task success across both interface conditions, with most participants correctly identifying the purpose of the website and locating the contact page. This indicates that both dark mode and light mode are effective in supporting usability and comprehension.

The qualitative data supports these findings by showing that the participants generally perceived the interface as clear, understandable and easy to navigate. Inspirational and informative as description of the website aligns with the high level of task success observed.

Furthermore, qualitative responses help explain minor variations in the data. The differences in brand recognition and interpretation suggest that while the overall design was effective, certain elements may not have been equally notable for all users.

Together, these findings indicate that while both interface modes perform similarly in measurable terms, user perceptions provide additional insights into how the design is experienced.

4.7 Result conclusion

By combining the quantitative and qualitative findings, a broader understanding of how interface themes influence readability and cognitive load can be established.

The quantitative data showed that both dark mode and light mode received high ratings for readability, usability, and task success. Subjective readability scores were slightly higher in dark mode, while cognitive load scores were similar across both conditions, indicating that neither theme created substantial mental strain.

The paired comparisons together with the correlation analysis of the likert-scale questions related to the two variables subjective readability and cognitive load further illustrated a moderate positive relationship between the two variables in both modes. Dark mode: ($r = 0.52$) and light mode: ($r = 0.49$).

These findings were supported by the qualitative responses, where participants frequently described both versions as clear, easy to navigate, and visually structured. Several participants associated dark mode with reduced eye strain and a calmer reading experience, while others described light mode as a sharper, more familiar and easier for sustained focus.

Although performance differences between conditions were small, the qualitative responses suggest that user preference plays an important role in perceived comfort. This indicates that cognitive load may be determined by individual habits, expectations, and visual preferences.

Overall, the findings suggest that both dark mode and light mode can effectively support content-heavy reading tasks when layout, spacing, contrast, and structure are well designed.

5 Discussion

The aim of this study is to investigate the relationship between subjective readability and cognitive load in the context of dark mode and light mode by combining qualitative and quantitative means of measurements in an within-subject experiment.

5.1 Research Question 1

> [1] How do dark and light interface themes influence subjective readability and cognitive load in content-heavy digital interfaces?

In relation to the first research question, the findings suggest that both dark mode and light mode can support subjective readability and low cognitive load in content-heavy digital interfaces when the interface is designed with clear structure, spacing, and accessible contrast ratios. The quantitative results showed only small differences between the two interface themes. Dark mode received somewhat higher mean scores for subjective readability ($M = 7.78$, $SD = 0.92$) as opposed to light mode ($M = 7.40$, $SD = 1.19$). When it comes to cognitive load, dark mode had higher scores ($M = 6.80$, $SD = 1.47$) than light mode ($M = 6.70$, $SD = 1.25$). Since the negative cognitive load items were reverse coded ahead of analysis, higher cognitive load scores signifies lower perceived cognitive load.

However, the paired t-tests that were conducted for both variables showed that these differences were not statistically significant. Subjective readability in dark and light mode had a p-value of .265 while cognitive load in dark and light mode had a p-value of .961. Both of these p-values surpassed the conventional significance threshold of .05. This demonstrates that both interface themes performed similarly in terms of readability and mental effort.

The standard deviation values obtained from the results provided additional insight into the participants' experiences. For subjective readability, responses were a bit more consistent in dark mode ($SD = 0.92$) compared to light mode ($SD = 1.19$), showing more consistency in agreement amongst participants for readability in the dark mode interface. For cognitive load in dark mode, the results were more variable ($SD = 1.47$) than light mode ($SD = 1.25$), revealing that participants had more mixed perceptions of mental effort when interacting with the dark mode interface. These variations in answers can be a result of personal experiences, familiarity with each interface theme, or personal preferences.

The differences in the results from the quantitative data were small and implies that neither color theme is objectively better based on the results from this study. Instead, one could argue that both themes have the possibility of good performance in the context of subjective readability and cognitive load when implemented together with good interface designs. This is further strengthened by Cognitive Load Theory that states that if a design is well thought out, it can reduce extraneous cognitive load and make it easier for users to process the information presented (Sweller, Ayres & Kalyuga, 2011). Consequently, it can be said that factors such as typography, contrast and layout may have more influence on subjective readability and cognitive load than the color theme alone.

The qualitative findings provide additional insights into how the participants experienced the two themes. Several participants described dark mode as “calm”, “soft for the eyes”, and less

visually straining during longer reading sessions. In contrast, the participants who preferred light mode often associated it with clarity, familiarity, and improved focus. These findings mean that subjective readability is both influenced by contrast and polarity, but also by individual visual preferences and reading habits.

The result can be connected to previous research regarding positive and negative polarity in digital interfaces. Earlier studies have suggested that light mode may improve readability due to higher perceived contrast and familiarity, while dark mode may reduce eye strain and improve visual comfort in extended use (Hall & Hanna, 2007). The findings of this study partially support both perspectives, as the participants reported positive reading experiences in both modes, but for different reasons.
From a cognitive load perspective, the findings indicate that neither interface theme introduced substantial extraneous load. Since the prototypes were identical in content, layout, typography, and structure, the differences in perceived mental effort can primarily be related to the visual presentation itself. The relatively high task success rate across both conditions further suggests that the interfaces supported efficient information processing without overloading working memory.

Overall, the results show that dark mode and light mode influence subjective readability and cognitive load in different but relatively balanced ways. Rather than one interface theme being objectively better, the findings suggest that well-structured design choices such as spacing, typography, and contrast may be more important for readability and cognitive comfort than the color theme alone.

5.2 Research Question 2

[2] How are these factors related during extended digital reading tasks?

The second research questions examined the relationship between subjective readability and cognitive load during extended digital reading tasks. The results from the correlation analysis showed a moderate positive relationship between the two variables in both interface conditions. In dark mode, the correlation coefficient was ($r = 0.52$), while light mode showed a similar correlation of ($r = 0.49$). This indicates that the participants who experienced the interface as easier to read also tended to report lower perceived cognitive load. This pattern can be seen illustrated in figure 6, which shows an upward trend between subjective readability and cognitive load, further reinforcing these results.

The coefficient of determination from the correlation analysis shows that subjective readability explains roughly 27.5% of the variance in cognitive load in dark mode ($R^2 = 0.275$) and 24.4% in light mode ($R^2 = 0.244$). These results imply that while subjective readability has a meaningful impact on users cognitive load, a considerable amount of variance is still undetermined. Based on this, it can be argued that other factors, such as the layout design, typography and personal experiences can also affect users' mental effort and subjective readability when interacting with content-heavy digital interfaces.

The qualitative findings further support this relationship. The participants who described the interface as easy to read or clear also frequently mentioned feeling less mentally tired and more focused while reading. This suggests that when the users perceive information as more accessible and well-structured, they also experience reduced cognitive load during reading.

However, the results also show that this relationship is not perfectly strong, meaning that subjective readability does not fully explain cognitive load. Even though there is a observable pattern of readability contributing to lower perceived mental effort, other surrounding factors could influence users cognitive load during digital tasks. Individual experiences and habits, familiarity with light and dark mode, personal preferences on visuals can all affect how mentally demanding an interface is perceived to be by the user.

These findings can be connected to cognitive load theory that explains that users working memory have limited resources when processing information (Paas, F., van Gog, T. & Sweller, J. 2010). When interfaces have clear visuals and easy navigation, it can reduce unnecessary extraneous load and users can focus their resources on understanding the content itself rather than being distracted by the designs. Therefore, the moderate relationship between cognitive load and subjective readability observed in this study is further supported by the theoretical claim that the two variables are connected somewhat, even though they can be influenced by other factors.

At the same time, it's important to remain cautious about the interpretation of the two variable relationships. Since the study relied on self reported data rather than objective performance measurements from participants, such as reading speed, the results can be influenced by the participants personal perceptions and may not fully reflect actual cognitive processing. Furthermore, the sample size in the study limits the generalizability of the findings due to it being quite small. In order to draw stronger conclusions about the relationship between the variables and its strength, a bigger sample size could increase the reliability of the statistical analysis and improve the possibility of generalizing the findings to a broader population.

In addition, the study did not include objective performance measurements such as reading speed, comprehension accuracy, or eye-tracking data. Therefore, the findings mainly reflect the participants' perceived experiences. Furthermore, the participants mainly consisted of university students familiar with digital reading environments, which may limit the generalizability of the findings to broader populations.

Despite these limitations, the study contributes to existing research by examining subjective readability and cognitive load simultaneously in the context of dark and light mode. Previous research has often focused on either objective performance or subjective preference separately, whereas this study indicates that the two factors appear moderately related during extended digital reading tasks.

5.3 Method discussion

This study used a mixed-method within-subject methodology in order to investigate how dark mode and light mode in web interfaces affect users' subjective readability and cognitive load. This approach was appropriate for this study since the purpose was not only to compare measurable differences in subjective readability in cognitive load, but to understand on a deeper level how the participants experienced the two interface themes (Creswell, 2014). The study investigated subjective experiences, user perceptions and preferences so the need for both qualitative and quantitative data was important. The quantitative survey results allowed comparison between dark mode and light mode while the qualitative responses provided deeper insights into each participant's personal perceptions.

An experimental design for this study was chosen since the aim of the study is to isolate dark mode and light mode and compare the two. The purpose was to see the effects on subjective readability and cognitive load for each interface color theme. By creating two prototypes, where variables such as content, typography and layout was controlled, the study could focus on the primary influence of the visual presentation of each color theme. The design further followed the Web Accessibility guidelines WCAG guidelines, more specifically Success Criterion 1.4.3 (Contrast (Minimum)) and Success Criterion 1.4.12 (Text Spacing) in order to decrease the influence of extraneous load caused by poor design choices. This method increased the ability to see if the differences in participants' experiences were related to the interface themes rather than other unnecessary design variables that would influence the results (Sweller, Ayres & Kalyuga, 2011).

Furthermore, since the aim of the was, as stated previously, to study subjective variables such as perceived readability, perceived mental effort and users visual comfort and user experience, self-reported measurements were seen as appropriate for this study. It captured participants' personal perceptions and subjective experiences. These concepts are strongly connected to individual interpretations and perceptions which is why it could be difficult to measure through only objective performance data.

The sample group consisting of university students are considered a relevant participant group for this study. These students in the ages of 19-30 frequently interact with content-heavy digital environments and their experiences makes them suitable participants in a study with the aim of investigating subjective readability and cognitive load in digital interfaces (Ahmed, 2024).

5.4 Methodological insights

The methodological approach in the study made it possible to combine quantitative and qualitative measurements and insights to gain a broader perspective and understanding of how the participants were affected by each mode. The measurable survey results together with qualitative insights provided a deeper explanation of participants' opinions and experiences. Even though the quantitative results indicated small differences between the two modes, the qualitative data explained personal preferences on a deeper level.

By having each participant interact with both prototypes, light mode and dark mode, it minimized individual differences and influences on the results such as visual preferences and previous experiences. Counterbalancing the order of the prototype exposure for the participants, i.e. half started with dark mode and the other half started with light mode, minimized potential order effect. This way reduces learning effects, fatigue effects and adaptation bias. It strengthened the internal validity of the study as well as fairness in comparison since participants were able to make a more direct comparison between each mode.

When it comes to the two prototypes, they were identical in content, typography, spacing, layout and structure. The only thing that differed were the color themes. Due to this, the study had controlled prototypes which lead to us having the ability to isolate the independent variable, the visual presentation, and use it as the primary influencing factors for the results.

Another strength in this methodological approach and the study is the use of accessibility-based design. Throughout the study, the WCAG guidelines, contrast ration standards, typography and spacing guidelines were followed to put focus on accessibility in the prototypes. This choice further improves the realism of the prototypes as well as ethical and usability qualities. Since the design of the prototypes were well thought out, the validity of the results were strengthened as well since poor scores in subjective readability and high mental load was less likely caused by bad design. The interface designs minimized unnecessary visual strain and extraneous load.

Furthermore, a strength in this study is the controlled testing environment that was created during the experiments. The experiments were conducted in a private room with controlled lighting, without external distraction and controlled exposure time to each prototypes. This controlled environment improved the study's consistency, reliability and the fairness between participants since they were exposed to similar conditions. It reduced external variables that could influence the participants' subjective readability and cognitive load that could harm the validity of the results.

5.5 Methodological limitations

Since the study had a relatively small sample size, 22 participants and 21 paired responses in the t-tests, it could serve as a limitation. The sample size reduced the statistical power of the analysis as well as the generalizability of the results. Moreover, it limits the ability to detect small differences in the results and makes correlations of the answers less reliable.

The study also used convenience sampling when conducting the recruitment of the participants. Even though the sampling is a justified choice, it can be seen as a limitation as well. University students were chosen with similar education, age and experiences in digital environments which may not represent a broader population. This can result in a sampling bias. Older users and other demographic groups with limited digital literacy and diverse accessibility needs are not fully reflected in the results and the findings may not reflect their perceived readability and cognitive load.

By relying on self-reported measurements of perceived readability and cognitive load, the findings in the study are reflecting participants personal perceptions and may not reflect actual performance. While subjective measures were needed and gave valuable understandings of user experience throughout the study, they cannot reflect task efficiency, accuracy or cognitive effort. Objective performance, eye tracking and psychological measures were not taken into account in the study. The absence of objective performance measurements restricts the ability to see whether perceived readability has a relationship to actual cognitive performance. This absence of objective measurements limits the ability to see whether there is an actual correlation between subjective readability, cognitive load and actual cognitive performance.

While the screen brightness was controlled throughout the majority of the experiments, minor variations may still have occurred due to limitations of keeping identical brightness levels across participants. Since the prototypes were presented on MacBooks, this could also present challenges in ensuring full control of the screen brightness. MacBook devices can sometimes automatically adjust the brightness levels of the screen based on ambient lighting conditions, and even though the lighting was controlled in the experiment environment, this should still be taken into account as a minor possible limitation to consider.

5.6 Reliability and validity reflection

The study demonstrates reasonable levels of reliability and validity due to the controlled experimental conditions and standardized procedure. All participants completed the same tasks, interacted with identical prototypes except for the color theme, and participated under similar environmental conditions. This increased the consistency of the data collection process and reduced the influence of external variables.
The within-subject design further strengthened the internal validity of the study since each participant interacted with both interface conditions. This reduced the impact of individual differences such as reading habits, visual preferences, and previous experiences with digital interfaces. Counterbalancing the order of prototype exposure also minimized potential learning and fatigue effects.

However, some limitations affecting validity and reliability remain. Since the study relied primarily on self-reported data, the participants may have interpreted questions differently or been influenced by temporary factors like concentration or fatigue. In addition, the relatively small and homogeneous sample limits the external validity of the findings, meaning that the results may not fully represent broader populations outside university students.

Nevertheless, the study provides valuable insights into the relationship between subjective readability and cognitive load in digital interfaces and demonstrates a methodologically structured approach to investigating user experience in dark and light mode environments.

5.7 Methodological improvements

Several methodological improvements could strengthen future studies on subjective readability and cognitive load in digital interfaces. One improvement would be to include a larger and more diverse participant group in order to increase the generalizability of the findings. Including participants of different ages, professional backgrounds, and levels of digital experience could provide a broader understanding of how interface themes are experienced across populations.

Another improvement would be to combine subjective self-reported measurements with objective performance-based methods such as reading speed analysis, comprehension tests, eye-tracking or physiological measurements. This could provide a more comprehensive understanding of how subjective readability relates to actual cognitive processing and performance.

Future studies could also investigate additional contextual factors that may influence, such as device type, ambient lighting conditions, screen brightness or prolonged exposure over longer periods of time. Since this study was conducted in a controlled indoor environment, a more naturalistic testing environment could contribute to a deeper understanding of real-world digital reading experiences.

Finally, future research could examine more variations of dark and light mode designs, including different contrast levels, typography choices, and customizable interface settings, in order to further investigate how visual presentation affects user experience and cognitive load.

6 Conclusions and further research

This chapter presents the overall conclusion of the study together with suggestions for future research within the field of dark mode, light mode, subjective readability, and cognitive load.

6.1 Conclusions

The purpose of this study was to investigate how dark mode and light mode in digital interfaces influence subjective readability and cognitive load in content-heavy environments, as well as examine the relationship between these variables during extended digital reading tasks. By using a mixed-method within subject design, the study combined quantitative measurements with qualitative feedback in order to gain a broader understanding of users' experience with both interface themes.

The findings showed that both dark mode and light mode supported high levels of subjective readability and relatively low perceived cognitive load. Although dark mode received slightly higher readability scores and slightly lower perceived cognitive load, the differences between the two interface themes were small and not statistically significant. This suggests that both themes can function effectively in content-heavy digital environments when the interface is designed with accessible contrast ratios, clear structure, appropriate spacing, and readable typography.

The qualitative findings further demonstrated that the participants associated dark mode with calmness, reduced eye strain and visual comfort, while light mode was often connected to clarity, familiarity, and improved focus. These findings indicate that user preferences and personal reading habits may influence how readability and mental effort are experienced.

In addition, the correlation analysis showed a moderate positive relationship between subjective readability and cognitive load in both interface conditions. The participants who perceived the interface as easier to read also tended to report lower levels of mental effort. This suggests that subjective readability and cognitive load are connected factors in digital reading environments, particularly during longer reading tasks.

Overall, the findings indicate that the effectiveness of dark mode and light mode may depend less on the color theme itself and more on the overall quality of the interface design. Well-structured layouts, accessible contrasts, and visually clear presentation appear to play an important role in supporting readability, reducing unnecessary cognitive load, and improving the user experience.

6.2 Further research

Future research could expand this study by including larger and more diverse participant groups in order to improve the generalizability of the findings. Since the current study mainly

involved university students familiar with digital environments, future studies could investigate how different age groups, digital habits, or accessibility needs influence subjective readability and cognitive load.

Another important direction for future research would be to combine subjective self-reported data with objective performance-based measurements such as reading speed, comprehension accuracy, eye-tracking or physiological measurements. This could provide a deeper understanding of how perceived readability relates to actual cognitive processing and task performance.

Future studies could also explore how different contextual factors affect user experience in dark mode and light mode, including device type, ambient lighting conditions, screen brightness, and prolonged exposure over time. Future research could also investigate different variations of interface themes, such as adjustable contrast levels, typography variations, or personalized visual settings, in order to better understand how visual presentation influences readability and cognitive load in digital interfaces.

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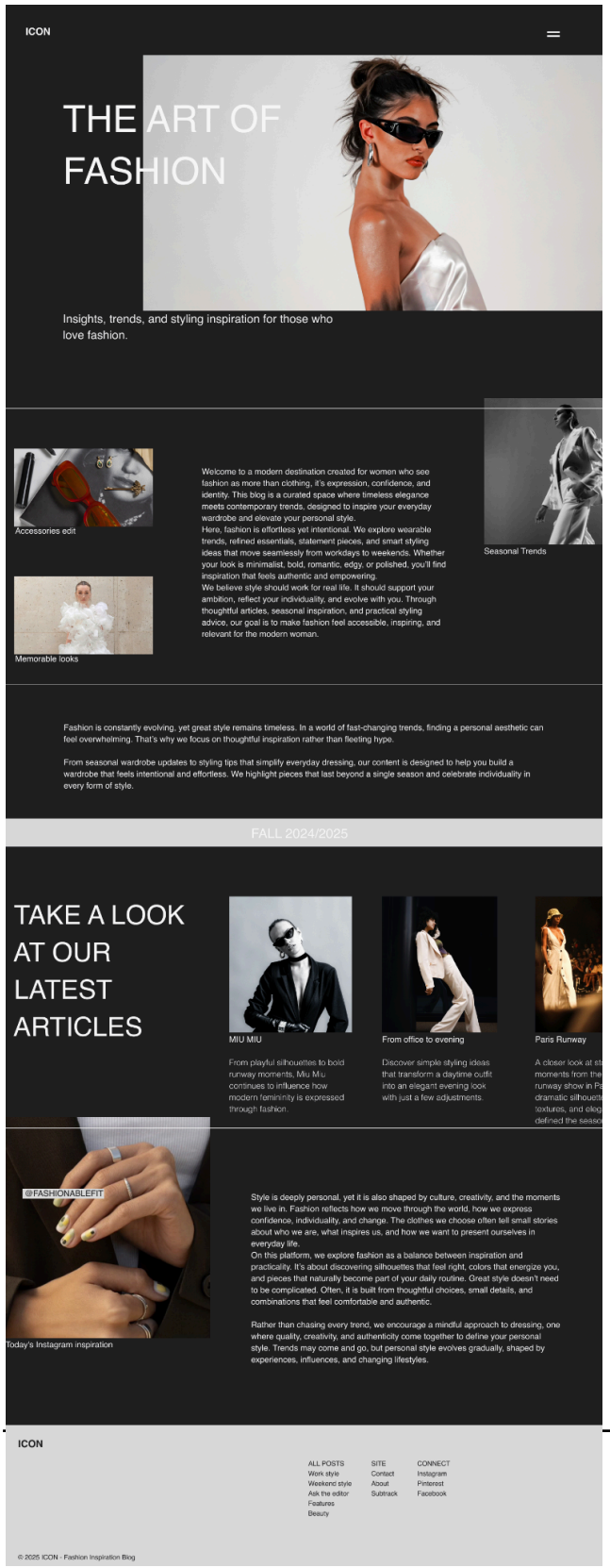
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8 Appendixes

Appendix A - Dark mode prototype

https://www.figma.com/proto/xp99cfnGrR4lRb2VjqIMi4/Untitled?node-id=4051-7&t=7eSfxXKuw1rTxvBk-1&scaling=min-zoom&content-scaling=fixed&page-id=4005%3A15



Appendix B - Light mode prototype

https://www.figma.com/proto/xp99cfnGrR4lRb2VjqIMi4/Untitled?node-id=4004-18&t=WjL8MFR9ZzHBHxS3-1&scaling=min-zoom&content-scaling=fixed&page-id=4005%3A15

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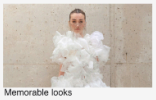
THE ART OF FASHION



Insights, trends, and styling inspiration for those who love fashion.

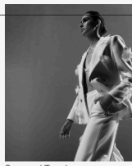


Accessories edit



Memorable looks

Welcome to a modern destination created for women who see fashion as more than clothing, it's expression, confidence, and identity. This blog is a curated space where timeless elegance meets contemporary trends, designed to inspire your everyday wardrobe and elevate your personal style.
Here, fashion is effortless yet intentional. We explore wearable trends, refined essentials, statement pieces, and smart styling ideas that move seamlessly from workdays to weekends. Whether your look is minimalist, bold, romantic, edgy, or polished, you'll find inspiration that feels authentic and empowering.
We believe style should work for real life. It should support your ambition, reflect your individuality, and evolve with you. Through thoughtful articles, seasonal inspiration, and practical styling advice, our goal is to make fashion feel accessible, inspiring, and relevant for the modern woman.



Seasonal Trends

Fashion is constantly evolving, yet great style remains timeless. In a world of fast-changing trends, finding a personal aesthetic can feel overwhelming. That's why we focus on thoughtful inspiration rather than fleeting hype.

From seasonal wardrobe updates to styling tips that simplify everyday dressing, our content is designed to help you build a wardrobe that feels intentional and effortless. We highlight pieces that last beyond a single season and celebrate individuality in every form of style.

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Paris Runway

A closer look at st
moments from the
runway show in Pa
dramatic silhouette
textures, and eleg
defined the seaso



Today's Instagram inspiration

Style is deeply personal, yet it is also shaped by culture, creativity, and the moments we live in. Fashion reflects how we move through the world, how we express confidence, individuality, and change. The clothes we choose often tell small stories about who we are, what inspires us, and how we want to present ourselves in everyday life.
On this platform, we explore fashion as a balance between inspiration and practicality. It's about discovering silhouettes that feel right, colors that energize you, and pieces that naturally become part of your daily routine. Great style doesn't need to be complicated. Often, it is built from thoughtful choices, small details, and combinations that feel comfortable and authentic.

Rather than chasing every trend, we encourage a mindful approach to dressing, one where quality, creativity, and authenticity come together to define your personal style. Trends may come and go, but personal style evolves gradually, shaped by experiences, influences, and changing lifestyles.

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Appendix C - Consent form

This survey is part of a thesis project investigating how dark mode and light mode in digital interfaces affect subjective readability and cognitive load.

You will be asked to view two interface prototypes and complete a set of tasks, followed by questions about your reading experience.

Your responses are anonymous and will only be used for research purposes. The survey takes approximately 5–7 minutes to complete.

Do you agree to participate in this study and allow your responses to be used for academic research?

- ☐ Yes
- ☐ No

Appendix D - Participants' Likert-scale responses for readability items in dark mode, including participant means, overall mean and standard deviation

Participants:	Q1	Q2	Q3	Q4	Q5	Mean	Overall mean:	SD:
P1	7	7	9	9	9	8,2		
P2	9	9	8	8	9	8,6		
P3	8	7	9	9	7	8		
P4	7	6	8	6	8	7		
P5	7	9	8	8	8	8		
P6	7	8	8	8	9	8		
P7	9	9	9	9	9	9		
P8	6	8	5	8	8	7		
P9	9	9	9	9	9	9		
P10	5	8	7	8	4	6,4		
P11	8	7	9	8	8	8		
P12	8	9	8	8	8	8,2		
P13	9	9	9	8	8	8,6		
P14	9	9	9	9	9	9		
P15	6	7	8	8	6	7		
P16	9	4	7	7	8	7		
P17	8	8	6	7	9	7,6		
P18	6	6	8	8	9	7,4		
P19	6	6	6	7	5	6		
P20	7	7	5	5	7	6,2		
P21	7	8	8	8	9	8		
P22	9	9	9	9	9	9	7,781818182	0,927688558

Appendix E - Participants' Likert-scale responses for cognitive load items in dark mode, including participant means, overall mean and standard deviation

Participants	Q1	Q2	Q3	Q5	Q5	Mean	Overall mean	SD
P1	6	8	8	2	6	6		
P2	9	9	9	9	9	9		
P3	5	6	8	2	7	5,6		
P4	7	3	8	4	7	5,8		
P5	9	7	8	3	7	6,8		
P6	6	7	8	2	7	6		
P7	8	8	8	9	9	8,4		
P8	4	7	7	6	5	5,8		
P9	9	8	9	9	9	8,8		
P10	2	9	4	2	9	5,2		
P11	4	8	8	2	8	6		
P12	8	9	9	8	9	8,6		
P13	9	9	9	9	9	9		
P14	6	6	6	4	4	5,2		
P15	7	7	8	8	8	7,6		
P16	2	6	8	3	8	5,4		
P17	7	7	7	7	8	7,2		
P18	8	5	3	8	8	6,4		
P19	5	5	5	3	4	4,4		
P20	8	8	8	8	8	8		
P21	4	8	8	3	7	6		
P22	9	9	9	9	9	9	6,827272727	1,47266581

Appendix F - Participants' Likert-scale responses for readability items in light mode, including participant means, overall mean and standard deviation

Particpants	Q1	Q2	Q3	Q4	Q5	Mean	Overall mean	SD
P1	6	6	6	5	4	5,4		
P2	7	9	6	7	7	7,2		
P3	9	8	8	8	9	8,4		
P4	6	8	7	9	7	7,4		
P5	3	4	7	4	7	5		
P6	7	9	9	9	6	8		
P7	9	9	9	9	9	9		
P8	6	8	9	8	8	7,8		
P9	9	9	9	8	9	8,8		
P10	8	8	8	8	8	8		
P11	9	5	7	6	6	6,6		
P12	7	7	8	9	7	7,6		
P13	8	7	9	8	9	8,2		
P14	9	8	7	7	9	8		
P15	9	8	8	8	9	8,4		
P16	5	7	8	8	7	7		
P17	8	8	7	8	8	7,8		
P18	3	6	4	5	6	4,8		
P19	6	5	6	8	8	6,6		
P20	7	6	6	7	8	6,8		
P21	9	9	9	8	8	8,6	7,4	1,191637529

Appendix G - Participants' Likert-scale responses for cognitive load items in light mode, including participant means, overall mean and standard deviation

Participants	Q1	Q2	Q3	Q4	Q5	Mean	Overall mean	SD
P1	8	5	8	4	7	6,4		
P2	9	9	8	8	9	8,6		
P3	7	7	9	3	7	6,6		
P4	9	2	9	4	7	6,2		
P5	4	6	4	4	7	5		
P6	9	9	8	9	8	8,6		
P7	9	9	7	9	9	8,6		
P8	8	8	7	2	9	6,8		
P9	9	8	8	8	8	8,2		
P10	8	8	8	2	8	6,8		
P11	5	8	7	7	7	6,8		
P12	5	5	5	4	4	4,6		
P13	8	8	8	8	7	7,8		
P14	9	8	9	2	2	6		
P15	8	8	8	2	8	6,8		
P16	8	7	5	8	8	7,2		
P17	4	6	6	3	5	4,8		
P18	4	6	4	7	8	5,8		
P19	8	7	8	2	8	6,6		
P20	6	7	5	4	4	5,2		
P21	8	8	8	9	8	8,2	6,742857143	1,257207791